

Course Code: CS3001	Course Name: Computer Networks
Instructor Names: Dr. Sufian Hameed, Dr. Farrukh Salim, Mr. Shoaib Raza, and Ms. Yusra Kaleem	
Student Roll No:	Section No:

Instructions:

- Return the question paper.
- **All questions must be answered in answer script and according to the sequence given in the question paper.**
- In case of any ambiguity, you may make an assumption. But your assumption should not contradict any statement in the question paper.

Time: 01 Hour

Max Marks: 50 Points

Question 1:

CLO#: 01 [10 Points]

Give short answers of the following.

- What is the conductor in fiber optic cable?
- Provide any three factors that affect the propagation of wireless signals.
- The same wire in coax cable is used, both for sending and receiving data, so number of downloads and uploads should be equal possible. But in reality, the downstream (download) rate is usually higher than upstream (upload) rate, how?
- Which wireless communication technology supports the highest data rate?
- Which physical wired medium supports the highest data rate?

Question 2:

CLO#: 01 [10 Points]

A user in Karachi, connected to the internet via a 2 Mbps connection retrieves a 25 Kbytes web page from a web server in Islamabad, where the page references 3 images of 200 Kbytes each. Assume that the one way propagation delay is 20 ms.

- How long does it take for the page (including images) to appear on the user's screen, assuming non persistent HTTP using a single connection at a time?
- What would become the time for the page (including images) to appear on the user's screen if he used persistent HTTP without pipelining in place of non-persistent HTTP?

Question 3:

CLO#: 02 [10 Points]

You send an e-mail message to your friend amir@gmail.com with a copy to your other friend marco@live.com. The subject of the message is "Pizza" and the text of the message is "Let's go out for a pizza tonight".

- Assuming your e-mail address is student@liu.edu.lb and that your mail user-agent is configured to use the mail server liu.edu.lb, write the complete SMTP exchange between your user agent and the mail server of your friends, including the whole body of the message.
- Name 2 protocols that your friends can use to access their inbox and read the mail message.
- What do you have to do in order to send a photo of the pizza attached to your mail message?

Question 4:

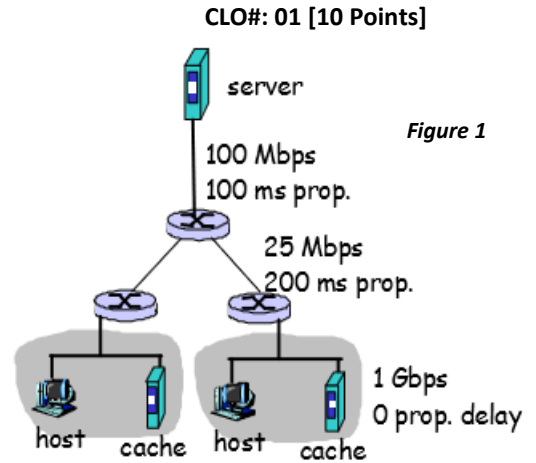
CLO#: 02 [10 Points]

You want to register your company "Securiteam" website (securiteam.org with IP address "10.0.0.1") and mail server (mail.securiteam.org with IP address "172.16.5.25") into the internet.

- What type of DNS server contains the records for the IP addresses of the website and the mail server of your company? Give the two corresponding resource records.
- If an internet user is searching for the company website's IP for the first time, what are in order, the DNS servers that the user's local DNS server need to query?

Question 5:

Consider the scenario in the Figure 1, in which a server is connected to a router by a 100 Mbps link, with a 100 ms propagation delay. That router in turn is connected to two routers, each over a 25 Mbps link with a 200 ms propagation delay. A Gbps link connects a host and a cache (when present) to each of these routers; this link, being a local area network, has a propagation delay that is essentially zero. All packets in the network are 10,000 bits long.



- i. What is the end-to-end delay from when a packet is transmitted by the server to when it is received at a host? Assume that there are no caches, that there is no queuing delay at a link, and that the node (router) packet-processing delays are also zero.
- ii. First assume that client hosts send requests for files directly to the server (i.e., the caches are off). What is the maximum rate at which the server can deliver data to a single client, assuming no other clients are making requests?
- iii. Again assume that only one client is active, but now suppose the caches are HTTP caches and are turned on. A client HTTP GET is always first directed to its local cache. 50% of the requests can be satisfied by the local cache. What is the maximum rate at which this client can receive data in this scenario?
- iv. Now suppose that the clients in both LANs are active and the HTTP caches are on, as in iii) above. 50% of the requests can be satisfied by the local cache. What is the maximum rate at which each client can receive data, in this scenario?
- v. Now suppose the 100 Mbps link is replaced by a 25 Mbps link. Repeat question iv) above in this new scenario.

Best of Luck!!!